

CHAPTER 10

Biotechnology and Its Applications

Biotechnological Applications in Agriculture

- The Bt toxin in genetically engineered Bt cotton kills the pest by: (2024 Re)
 - creating pores in the midgut.
 - damaging the respiratory system.
 - degenerating the nervous system.
 - altering the pH of body fluids.
- Following are the steps involved in action of toxin in Bt Cotton. (2024 Re)
 - The inactive toxin converted into active form due to alkaline pH of gut of insect.
 - Bacillus thuringiensis* produce crystals with toxic insecticidal proteins.
 - The alkaline pH solubilises the crystals.
 - The activated toxin binds to the surface of midgut cells, creates pores and causes death of the insect.
 - The toxin proteins exist as inactive protoxins in bacteria.
 Choose the **correct** sequence of steps from the options given below:
 - E→C→B→A→D
 - B→C→A→E→D
 - A→E→B→D→C
 - B→E→C→A→D
- Given below are two statements: (2024 Re)

Statement I: RNA interference takes place in all Eukaryotic organisms as method of cellular defense.

Statement II: RNAi involves the silencing of a specific mRNA due to a complementary single stranded RNA molecule that binds and prevents translation of mRNA

 In the light of the above statements, choose the correct answer from the options given below.
 - Statement I is true but Statement II is false.
 - Statement I is false but Statement II is true.
 - Both Statement I and Statement II are true.
 - Both Statement I and Statement II are false.
- The capacity to generate a whole plant from any cell of the plant is called: (2024)
 - Differentiation
 - Somatic hybridization
 - Totipotency
 - Micropropagation
- Given below are two statements: (2024)

Statement I: Bt toxins are insect group specific and coded by a gene *cry IAc*.

Statement II: Bt toxin exists as inactive protoxin in *B. thuringiensis*. However, after ingestion by the insect the inactive protoxin gets converted into active form due to acidic pH of the insect gut.

 In the light of the above statements, choose the correct answer from the options given below:
 - Statement I is true but Statement II is false
 - Statement I is false but Statement II is true
 - Both Statement I and statement II are true
 - Both Statement I and Statement II are false
- Which of the following are fused in somatic hybridization involving two varieties of plants? (2024)
 - Protoplasts
 - Pollens
 - Callus
 - Somatic embryos
- Which of the following statement is **incorrect** about *Agrobacterium tumifaciens*? (2023-Manipur)
 - It is used to deliver gene of interest in both prokaryotic as well as eukaryotic host cells.
 - 'Ti' plasmid from *Agrobacterium tumifaciens* used for gene transfer is not pathogenic to plant cell.
 - It transforms normal plant cells into tumor cells.
 - It delivers 'T-DNA' into plant cell.
- Transposons can be used during which one of the following? (2022)
 - Gene sequencing
 - Polymerase Chain Reaction
 - Gene silencing
 - Autoradiography
- Bt cotton variety that was developed by the introduction of toxin gene of *Bacillus thuringiensis* (Bt) is resistant to (2020)
 - Fungal diseases
 - Plant nematodes
 - Insect predators
 - Insect pests
- RNA interference is used for which of the following purposes in the field of biotechnology? (2020-Covid)
 - To develop a pest resistant plant against infestation by nematode
 - To enhance the mineral usage by the plant
 - To reduce post harvest losses
 - To develop a plant tolerant to abiotic stresses

11. Which of the following is true for Golden rice? (2019)
- It is Vitamin A enriched, with a gene from daffodil
 - It is pest resistant, with a gene from *Bacillus thuringiensis*
 - It is drought tolerant, developed using *Agrobacterium* vector
 - It has yellow grains, because of a gene introduced from a primitive variety of rice
12. What triggers activation of protoxin to active Bt toxin of *Bacillus thuringiensis* in bollworm? (2019)
- Body temperature
 - Moist surface of midgut
 - Alkaline pH of gut
 - Acidic pH of stomach
13. Which part of the tobacco plant is infected by *Meloidogyne incognita*? (2016 - I)
- Flower
 - Leaf
 - Stem
 - Root
14. In Bt cotton, the Bt toxin present in plant tissue as pro-toxin is converted into active toxin due to: (2015)
- Action of gut micro-organism
 - Presence of conversion factors in insect gut
 - Alkaline pH of the insect gut
 - Acidic pH of the insect gut
15. Golden rice is a genetically modified crop plant where the incorporated gene is meant for biosynthesis of: (2015 Re)
- Vitamin C
 - Omega 3
 - Vitamin A
 - Vitamin B
16. Which of the following Bt crops is being grown in India by the farmers? (2013)
- Soyabean
 - Maize
 - Cotton
 - Brinjal

Biotechnological Applications in Medicine

17. Match List-I with List-II: (2024 Re)

List-I		List-II	
(A)	Genetically engineered Human Insulin	(I)	Gene therapy
(B)	GM Cotton	(II)	<i>E.coli</i>
(C)	ADA Deficiency	(III)	Antigen-antibody interaction
(D)	ELISA	(IV)	<i>Bacillus thuringiensis</i>

Choose the **correct** answer from the options given below:

- A-III, B-II, C-IV, D-I
 - A-II, B-I, C-IV, D-III
 - A-IV, B-III, C-I, D-II
 - A-II, B-IV, C-I, D-III
18. Which one of the following techniques does not serve the purpose of early diagnosis of a disease for its early treatment? (2023)
- Enzyme Linked Immuno-Sorbent Assay (ELISA) technique
 - Recombinant DNA Technology
 - Serum and Urine analysis
 - Polymerase Chain Reaction (PCR) technique

19. Select **incorrect** statement, regarding chemical structure of insulin. (2023-Manipur)

- Mature insulin molecule consists of three polypeptide chains-A, B and C.
- Insulin is synthesized as prohormone which contains extra stretch of C-peptide.
- C-peptide is not present in mature insulin molecule.
- Polypeptide chains A and B are linked by disulphide bridges.

20. Match List-I with List-II. (2023-Manipur)

List-I		List-II	
(A)	Gene therapy	(I)	Separation of DNA fragments
(B)	RNA interference	(II)	Diagnostic test for AIDS
(C)	ELISA	(III)	Cellular defence
(D)	Gel Electrophoresis	(IV)	Allows correction of a gene defect

Choose the **correct** answer from the options given below:

- (A)-(IV), (B)-(I), (C)-(II), (D)-(III)
- (A)-(IV), (B)-(II), (C)-(III), (D)-(I)
- (A)-(IV), (B)-(III), (C)-(II), (D)-(I)
- (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

21. Match List-I with List-II: (2022)

List-I		List-II	
(A)	Gene gun	(I)	Replacement of a faulty gene by a normal healthy gene
(B)	Gene therapy	(II)	Used for transfer of Gene
(C)	Gene cloning	(III)	Total DNA in the cells of an organism
(D)	Genome	(IV)	To obtain identical copies of a particular DNA molecule

Choose the correct answer from the options given below:

- A-II, B-III, C-IV, D-I
 - A-II, B-I, C-IV, D-III
 - A-I, B-III, C-II, D-IV
 - A-IV, B-I, C-III, D-II
22. In gene therapy of Adenosine Deaminase (ADA) deficiency, the patient requires periodic infusion genetically engineered lymphocytes because: (2022)
- Genetically engineered lymphocytes are not immortal cells.
 - Retroviral vector is introduced into these lymphocytes.
 - Gene isolated from marrow cells producing ADA is introduced into cells at embryonic stages
 - Lymphocytes from patient's blood are grown in culture, outside the body.

23. Statements related to human Insulin are given below. Which statement(s) is/are correct about genetically engineered Insulin? (2022)

A. Pro-hormone insulin contain extra stretch of C-peptide
 B. A-peptide and B-peptide chains of insulin were produced separately in *E.coli*, extracted and combined by creating disulphide bond between them.
 C. Insulin used for treating Diabetes was extracted from Cattles and Pigs.
 D. Pro-hormone Insulin needs to be processed for converting into a mature and functional hormone.
 E. Some patients develop allergic reactions to the foreign insulin.

Choose the most appropriate answer from the options given below.

- a. C, D and E only b. A, B and D only
 c. B only d. C and D only
24. When gene targeting involving gene amplification is attempted in an individual's tissue to treat disease, it is known as: (2021)
- a. Gene therapy b. Molecular diagnosis
 c. Safety testing d. Biopiracy
25. Which of the following is not an application of PCR (Polymerase Chain Reaction)? (2021)
- a. Gene amplification
 b. Purification of isolated protein
 c. Detection of gene mutation
 d. Molecular diagnosis
26. Which of the following is a correct sequence of steps in a PCR (Polymerase Chain Reaction)? (2021)
- a. Denaturation, Extension, Annealing
 b. Extension, Denaturation, Annealing
 c. Annealing, Denaturation, Extension
 d. Denaturation, Annealing, Extension
27. Now a days it is possible to detect the mutated gene causing cancer by allowing radioactive probe to hybridise its complementary DNA in a clone of cells, followed by its detection using autoradiography because: (2021)
- a. Mutated gene completely and clearly appears on a photographic film.
 b. Mutated gene does not appear on a photographic film as the probe has no complementarity with it.
 c. Mutated gene does not appear on photographic film as the probe has complementarity with it.
 d. Mutated gene partially appears on a photographic film.
28. With regard to insulin choose correct options. (2021)
- A. C-peptide is not present in mature insulin.
 B. The insulin produced by rDNA technology has C-peptide.
 C. The pro-insulin has C-peptide.
 D. A-peptide and B-peptide of insulin are interconnected by disulphide bridges.

Choose the correct answer from the options given below.

- a. B and C only b. A, C and D only
 c. A and D only d. B and D only

29. For effective treatment of the disease, early diagnosis and understanding its pathophysiology is very important. Which of the following molecular diagnostic techniques is very useful for early detection? (2021)

a. Southern Blotting Technique
 b. ELISA Technique
 c. Hybridization Technique
 d. Western Blotting Technique

30. The adenosine deaminase deficiency results into: (2021)

a. Parkinson's disease
 b. Digestive disorder
 c. Addison's disease
 d. Dysfunction of Immune system

31. Which of the following statements is not correct? (2020)

a. The proinsulin has an extra peptide called C-peptide
 b. The functional insulin has A and B chains linked together by hydrogen bonds
 c. Genetically engineered insulin is produced in *E.coli*.
 d. In man insulin is synthesised as a proinsulin.

32. Match the following columns and select the correct option (2020)

	Column - I		Column - II
1.	Bt cotton	(i)	Gene therapy
2.	Adenosine deaminase deficiency	(ii)	Cellular defence
3.	RNAi	(iii)	Detection of HIV infection
4.	PCR	(iv)	<i>Bacillus thuringiensis</i>

- (1) (2) (3) (4)
 a. (iii) (ii) (i) (iv)
 b. (ii) (iii) (iv) (i)
 c. (i) (ii) (iii) (iv)
 d. (iv) (i) (ii) (iii)

33. Which of the following is commonly used as a vector for introducing a DNA fragment in human lymphocytes? (2018)

a. Retrovirus b. Ti plasmid
 c. λ phage d. pBR322

34. Which kind of therapy was given in 1990 to a four year old girl with adenosine deaminase (ADA) deficiency? (2016 - II)

a. Immunotherapy b. Radiation therapy
 c. Gene therapy d. Chemotherapy

35. The two polypeptides of human insulin are linked together by: (2016 - I)

a. Hydrogen bonds b. Phosphodiester bond
 c. Covalent bond d. Disulphide bridges

36. ADA is an enzyme which is deficient in a genetic disorder SCID. What is the full form of ADA? (2014)

a. Adenosine DeoxyAminase
 b. Adenosine Deaminase
 c. Aspartate Deaminase
 d. Arginine Deaminase

37. The first human hormone produced by recombinant DNA technology is: (2014)

- a. Progesterone b. Insulin
c. Estrogen d. Thyroxin

Transgenic Animals

38. Match List-I with List-II: (2024)

List-I		List-II	
(A)	α - 1 antitrypsin	(I)	Cotton bollworm
(B)	<i>Cry IAb</i>	(II)	ADA deficiency
(C)	<i>Cry IAc</i>	(III)	Emphysema
(D)	Enzyme replacement therapy	(IV)	Corn borer

Choose the correct answer from the options given below :

- a. A-(III), B-(IV), C-(I), D-(II)
b. A-(II), B-(IV), C-(I), D-(III)
c. A-(II), B-(I), C-(IV), D-(III)
d. A-(III), B-(I), C-(II), D-(IV)
39. Milk of transgenic 'Cow Rosie' was nutritionally more balanced product for human babies than natural cow milk because it contained: (2022 Re)
- a. Human enzyme Adenosine Deaminase (ADA)
b. Human protein α -1-antitrypsin
c. Human alpha-lactalbumin
d. Human insulin-like growth factor

Ethical Issues

40. Given below are two statements: (2024 Re)

Statement I: The Indian Government has set up GEAC, which will make decisions regarding the validity of GM research.

Statement II: Biopiracy is the term used to refer to the use of bio-resources by native people.

In the light of the above statements, choose the correct answer from the options given below:

- a. Statement I is true but Statement II is false
b. Statement I is false but Statement II is true
c. Both Statement I and Statement II are true
d. Both Statement I and Statement II are false
41. The laws and rules to prevent unauthorised exploitation of bio-resources are termed as- (2020-Covid)
- a. Bioethics b. Bioengineering
c. Biopiracy d. Biopatenting
42. In India, the organisation responsible for assessing the safety of introducing genetically modified organisms for public use is: (2018)
- a. Indian Council of Medical Research (ICMR)
b. Council for Scientific and Industrial Research (CSIR)
c. Research Committee on Genetic Manipulation (RCGM)
d. Genetic Engineering Appraisal Committee (GEAC)
43. A 'new' variety of rice was patented by a foreign company though such varieties have been present in India for a long time. This is related to: (2018)
- a. Co-667 b. Sharbati Sonora
c. Lerma Rojo d. Basmati
44. Use of bioresources by multinational companies and organisations without authorisation from the concerned country and its people is called: (2018)
- a. Bio-infringement b. Biopiracy
c. Biodegradation d. Bioexploitation
45. Which body of the Government of India regulates GM research and safety of introducing GM organisms for public services? (2015)
- a. Genetic Engineering Approval Committee
b. Research Committee on Genetic Manipulation
c. Bio-safety committee
d. Indian Council of Agricultural Research

Answer Key

1. (a) 2. (d) 3. (a) 4. (c) 5. (a) 6. (a) 7. (a) 8. (c) 9. (d) 10. (a)
11. (a) 12. (c) 13. (d) 14. (c) 15. (c) 16. (c) 17. (d) 18. (c) 19. (a) 20. (c)
21. (b) 22. (a) 23. (c) 24. (a) 25. (b) 26. (d) 27. (b) 28. (b) 29. (b) 30. (d)
31. (b) 32. (d) 33. (a) 34. (c) 35. (d) 36. (b) 37. (b) 38. (a) 39. (c) 40. (a)
41. (d) 42. (d) 43. (d) 44. (b) 45. (a)